From tests to models

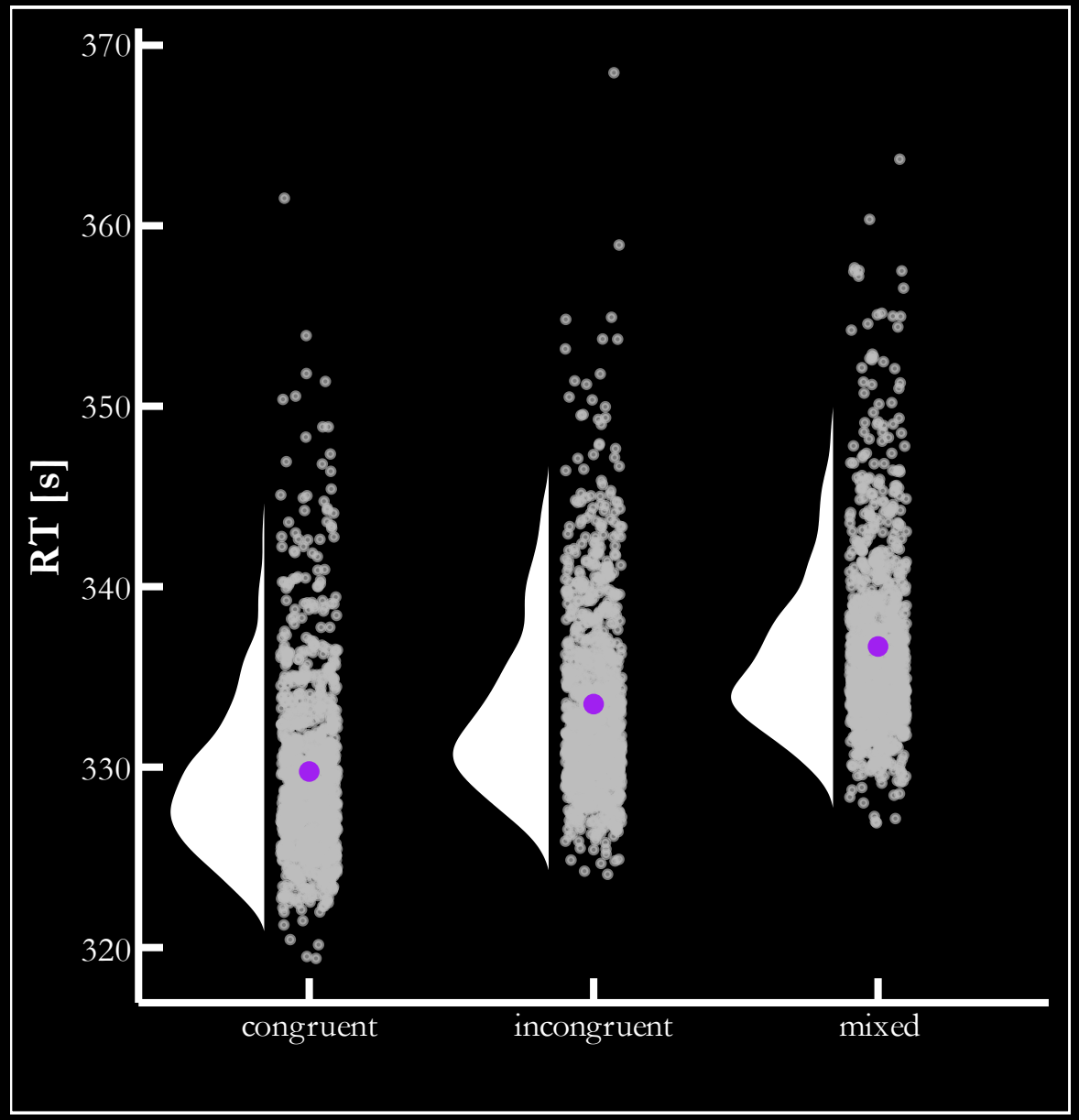
Objectives

- Demystifying data modelling
- Showing that modelling is more understandable than tests
- Showing that we often model without knowing it
- Moving to Bayesian models

Approach

- Unpacking a simple ANOVA
- Showing that ANOVA is a linear model* in disguise
- Running a first Bayesian model

* Technically is a method applied to a linear model



Are the means different?

How can we answer this question?

Frequentist	ANOVA Linear Moel
Bayesian	General Linear Model

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Argument:

a) ANOVA and Linear Model are equivalent
b) Linear models are more flexible than standard tests
c) Bayesian models offer more information than frequentists linear models
d) Then: we should opt for Bayesian models

Steps

1. Load data
2. Compute participants' and conditions' averages
3. Compute Variance Metrics
4. Get F-statistics
5. Working on p-value

NOTE: The code needs to be adjusted (do not rush through)

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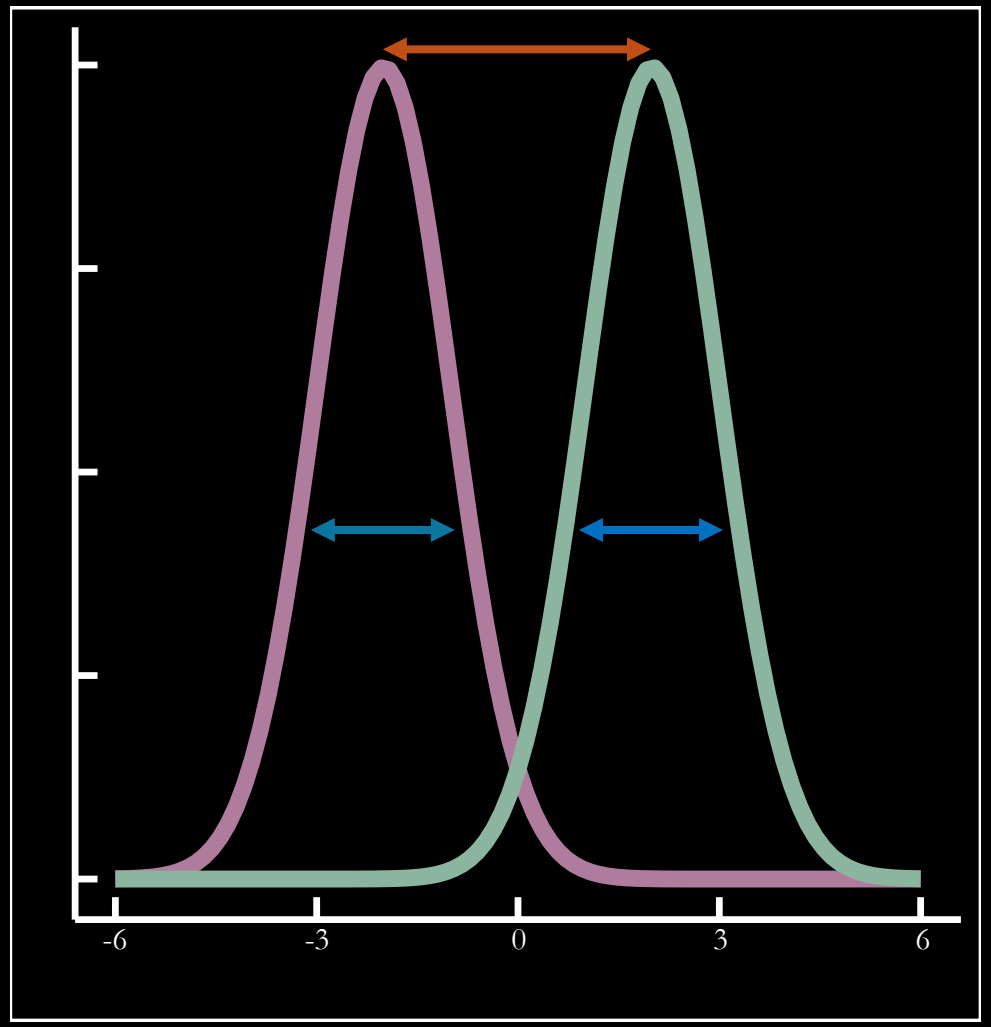

If variance between groups is higher than variance within groups, then the observations are likely drawn from two different distributions

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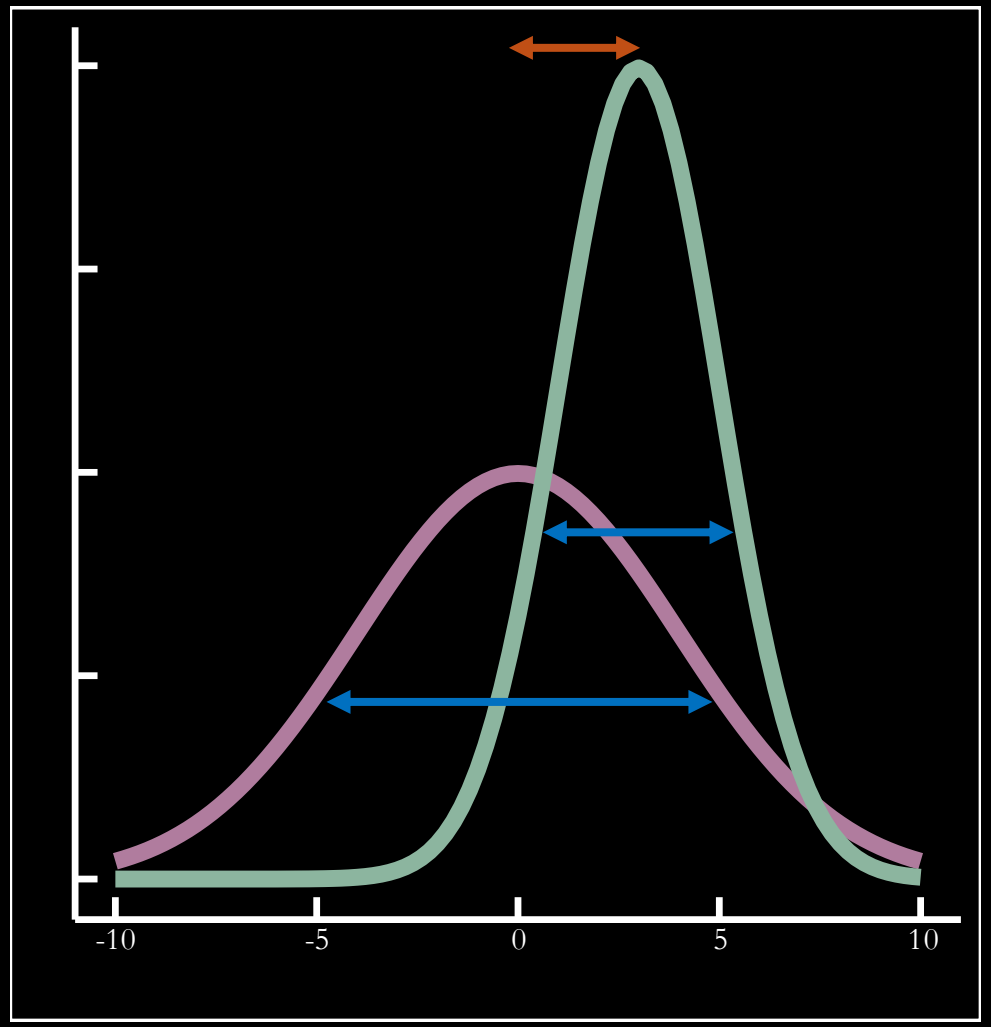

Otherwise, it's **hard to tell**

NOTE: This has implications for interpreting a null result!

F-value

$$F = \frac{Between\ Group\ Variability}{Within\ Group\ Variability} = \frac{\frac{SS_{between}}{Df_{between}}}{\frac{SS_{within}}{Df_{within}}} = \frac{SS_{between}}{Df_{between}} \times \frac{Df_{within}}{SS_{within}} = \frac{SS_{between}}{SS_{within}} \times \frac{Df_{within}}{Df_{between}}$$

SS = Sum of Squares

In case

$$SS_{within} = \sum (\overline{RT}_p - \overline{RT}_{g_p})^2$$

$$SS_{between} = \sum N_g \times (\overline{RT}_g - \overline{RT})^2$$

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ANOVA assumption!

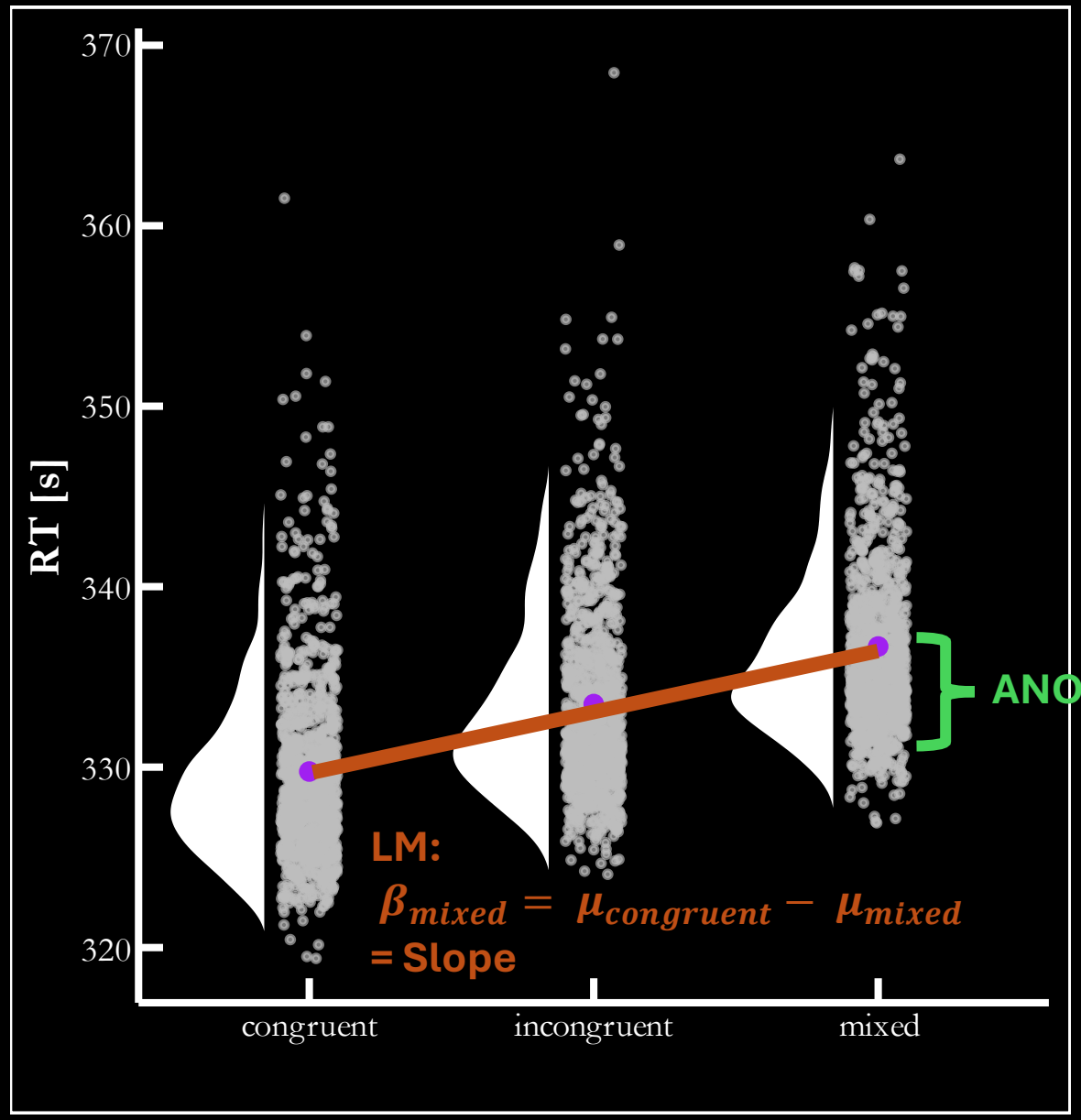
Linear Model

- H0 looks like a linear model, right?
  Indeed, ANOVA is a method **applied to** a linear model
- *F* has some similarities with R^2

$$F = \frac{SS_{between}}{SS_{within}} \times \frac{Df_{within}}{Df_{between}}$$

$$R^2 = 1 - \frac{SS_{residuals}}{SS_{total}}$$

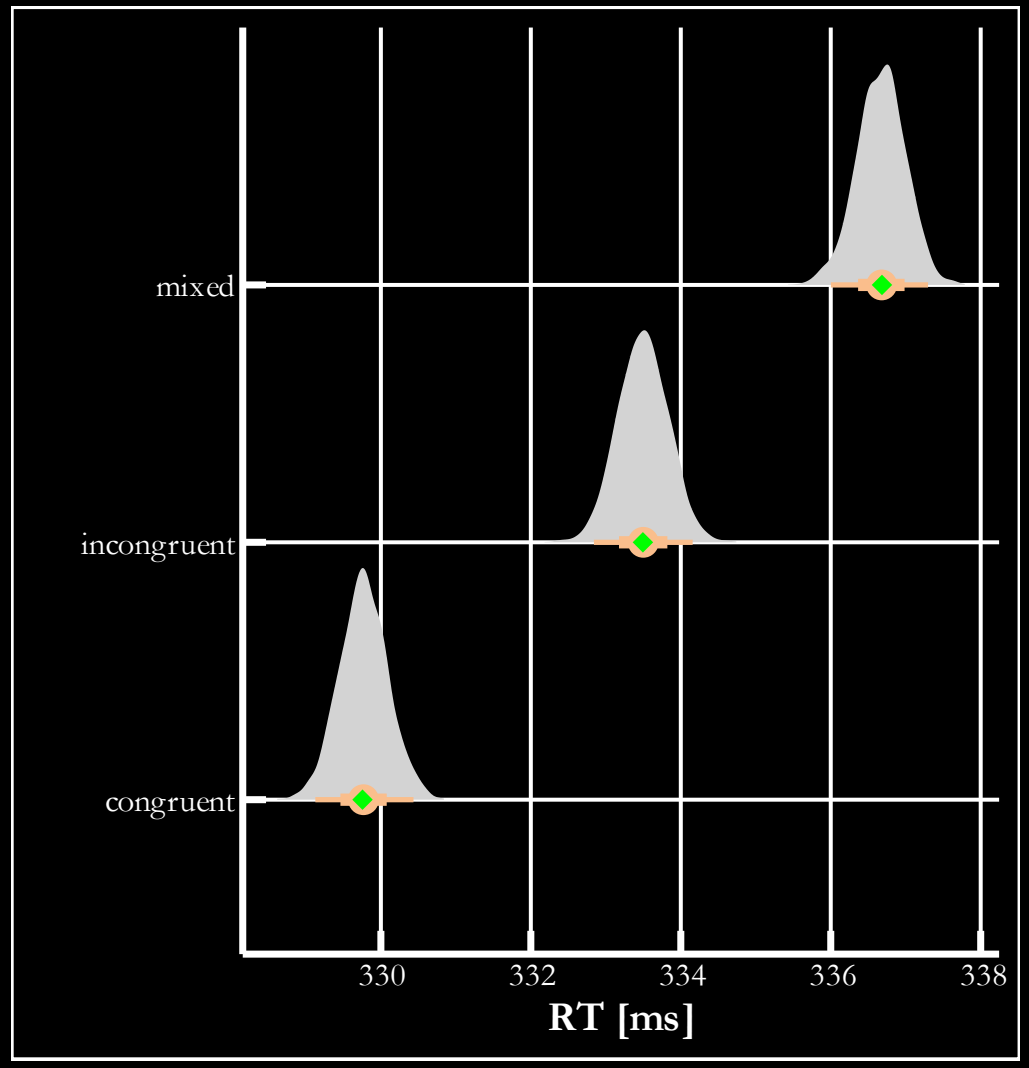
Proportion of variance in the outcome captured by the predictor


370
360
350
340
330
320
RT [s]
congruent
incongruent
mixed
LM:
$\beta_{mixed} = \mu_{congruent} - \mu_{mixed}$
= Slope
ANOVA: $\mu_{congruent} - \mu_{mixed}$

Going Bayesian

- A linear model is a more flexible tool than and ANOVA (or any other tests)
- The switch between the two it’s quite easy
- The switch to a Bayesian approach is quite easy too
  - In terms of code
  - As for anything, some aspects require learning and adjusting, BUT…
  - The benefits in terms of interpretability and general practice outweigh the challenges

Going Bayesian

- Both Frequentist and Bayesian approaches use likelihoods
  - We'll see the similarity through code

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